

Charles Wang

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EDUCATION

Purdue University M.S., Agricultural & Biological Engineering B.S., Biological Engineering GPA: 3.63	West Lafayette, IN Expected May 2027 Dec 2025
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EXPERIENCE

LeafSpec LLC / Purdue ABE Plant Sensor Lab Product Engineer <i>Purdue startup that builds handheld hyperspectral leaf imagers for plant scientists and breeders.</i> - Design and build handheld plant-imaging instruments: mechanical design in SolidWorks, FDM 3D printing, and integration of microcontrollers, motors, LED lighting, and cameras into working prototypes. - Collected the hyperspectral and multispectral leaf data for a peer-reviewed study on herbicide crop injury; the dataset trained machine learning models that reached 90% classification accuracy. - Work directly with agronomy and weed science researchers to define device requirements, including corn leaf scans in under 10 seconds and GPS/GIS tagging that maps each scan to its field location.	May 2024 – Present
Institute for Digital Forestry, Purdue University Undergraduate Summer Researcher <i>Paid summer research program, placed in the ABE Plant Sensor Lab.</i> - Rebuilt an undocumented multispectral leaf imager (UV to IR) using only legacy SolidWorks files: reverse-engineered the assembly, 3D-printed parts, and wired the camera, microcontroller, and drive electronics. - Redesigned the top enclosure to fit the microcontroller and driver board inside the main housing, replacing an external tethered box and shrinking total device volume by roughly 30%. - Verified image quality with a 0.04 mm/pixel camera that resolves individual leaf veins.	May – Jul 2024

PROJECTS

Leaf Cross-Section Microscope Imager M.S. Project, Plant Sensor Lab - Building a modular attachment for the lab's handheld 400× leaf microscope that images full corn-leaf cross sections for nitrogen sensing. - Designed a motor-driven clamp and stage that moves a leaf laterally through the optical path, letting the microscope capture the cross section frame by frame at 200×. - Added an electrically tunable liquid lens to the optical path, replacing a motor-and-gear focusing stage that was too slow and unstable for cross-section work. - Built a web-based control console (lighting, motor motion, live view, capture, labeling) to replace the old terminal workflow, and a desktop app that stitches frames into one image with manual alignment, color correction, and side-by-side comparison.	Jan 2026 – Present
LeafDoc, Smartphone Phenotyping Attachment Plant Sensor Lab - Led a team of 6 engineers designing a low-cost smartphone attachment for leaf-level plant phenotyping. - Designed a light box that blocks ambient light and a slide mechanism that flattens curled corn leaves for consistent capture. - Built the Android app and cloud pipeline: guided capture, image stitching, and LLM-based nutrient and disease detection that returns a plant health score.	Jun 2025 – Present
FluoroScan, Chlorophyll Fluorescence Imager Plant Sensor Lab - Engineered a fluorescence imager that measures photosynthesis (PSII) efficiency for under 1% of the cost of commercial instruments. - Built a mount with twelve 450 nm LEDs delivering 4,000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ of saturating light; captured 680 nm emission at 64% agreement with a reference instrument in initial testing.	Sep 2024 – May 2025

SKILLS

CAD & Prototyping: SolidWorks (parts, assemblies, enclosures), FDM 3D printing, hardware assembly and test
Imaging & Optics: Hyperspectral, multispectral, and fluorescence imaging; microscopy; liquid lens focusing; LED illumination
Electronics: Microcontroller integration, motor control, sensor wiring and cable management
Software: Python, C, MATLAB; image processing and stitching; web control interfaces; Android development; LLM API integration
Languages: English, Chinese (fluent)

HONORS & LEADERSHIP

- Top 10, Indiana Soybean Innovation Competition (2024): co-created Soysheets, a 67% soy detergent and dryer sheet. - Executive Board, Association of Hip Hop Practitioners & Scholars (2021–Present): raised \$80,000 through the “Hip Hop, the Next Giant Leap” fundraiser for performances and academic conferences at Purdue.
